JavaScript Fundamentals

Course Information

Course Name: JavaScript Fundamentals

Level: Beginner

Prerequisites:

- Basic computer knowledge
- Basic HTML and CSS (recommended but not required)

Estimated Duration: 10–12 Hours

Course Description

JavaScript is one of the most popular programming languages used to create interactive websites and web applications. In this course, students will learn JavaScript from scratch, including variables, operators, functions, loops, arrays, objects, DOM manipulation, and asynchronous programming. By the end of the course, students will be able to build small interactive web applications.

Learning Outcomes

After completing this course, students will be able to:

- Understand JavaScript syntax.
- Declare and use variables.
- Perform calculations using operators.
- Write conditional statements.
- Create loops.
- Write reusable functions.
- Work with arrays and objects.
- Manipulate web pages using the DOM.
- Handle events.
- Fetch data from APIs using asynchronous JavaScript.
- Build simple JavaScript applications.

Lesson 1: Introduction to JavaScript

Learning Objectives

After this lesson, students will be able to:

- Explain what JavaScript is.
- Understand where JavaScript runs.
- Write their first JavaScript program.

What is JavaScript?

JavaScript is a programming language used to make websites interactive.

Examples include:

- Image sliders
- Login validation
- Online calculators
- Games
- Dynamic menus
- Chat applications

JavaScript can run:

- In web browsers
- On servers using Node.js

Adding JavaScript

Inline

```
<button onclick="alert('Hello!')">Click</button>
```

Internal

```
<script>
console.log("Hello World");
</script>
```

External

```
<script src="app.js"></script>
```

Your First Program

```
console.log("Hello JavaScript");
```

Output

```
Hello JavaScript
```

Summary

- JavaScript adds interactivity.
- It works with HTML and CSS.
- Programs are executed line by line.

Lesson 2: Variables and Data Types

Learning Objectives

Students will learn:

- Variables
- Data types
- Naming rules

Variables

Variables store information.

```
let name = "Ali";
let age = 20;
```

Variable Types

let

Can change later.

```
let city = "Lahore";
city = "Islamabad";
```

---

const

Cannot be changed.

```
const PI = 3.14159;
```

---

var

Older method.

```
var score = 90;
```

Data Types

String

```
let name = "John";
```

Number

```
let age = 22;
```

Boolean

```
let isStudent = true;
```

Undefined

```
let x;
```

Null

```
let data = null;
```

Summary

Variables store values.

JavaScript has multiple data types.

Lesson 3: Operators

Arithmetic Operators

```
let a = 10;
let b = 5;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
console.log(a % b);
```

Comparison Operators

```
10 > 5
10 < 5
10 == "10"
10 === "10"
```

Explain the difference between `==` and `===`.

Logical Operators

```
&&
||
!
```

Example

```
let age = 20;

if(age >=18 && age<=30){
```

```
    console.log("Eligible");
}
```

Lesson 4: Conditional Statements

if Statement

```
let age = 20;

if(age >=18){
    console.log("Adult");
}
```

---

if...else

```
if(age>=18){
   console.log("Adult");
}
else{
   console.log("Minor");
}
```

---

else if

```
let marks=85;

if(marks>=90){
   console.log("A");
}
else if(marks>=80){
   console.log("B");
}
else{
   console.log("C");
}
```

---

switch

```
let day=2;

switch(day){
case 1:
console.log("Monday");
break;

case 2:
console.log("Tuesday");
break;

default:
console.log("Unknown");
}
```

Lesson 5: Loops

for Loop

```
for(let i=1;i<=5;i++){
console.log(i);
}
```

---

while Loop

```
let i=1;

while(i<=5){
console.log(i);
i++;
}
```

---

do...while

```
let i=1;

do{
console.log(i);
```

```
i++;
}
while(i<=5);
```

Lesson 6: Functions

Function Declaration

```
function greet(){
console.log("Welcome");
}

greet();
```

Function with Parameters

```
function add(a,b){
return a+b;
}

console.log(add(5,3));
```

Arrow Function

```
const square=(x)=>{
return x*x;
}
```

Lesson 7: Arrays

```
let fruits=["Apple","Banana","Orange"];
```

Access

```
console.log(fruits[0]);
```

Add

```
fruits.push("Mango");
```

Remove

```
fruits.pop();
```

Loop

```
for(let fruit of fruits){
console.log(fruit);
}
```

Lesson 8: Objects

```
let student={
name:"Ali",
age:20,
city:"Lahore"
};
```

Access

```
console.log(student.name);
```

Add Property

```
student.cgpa=3.8;
```

Lesson 9: DOM Manipulation

HTML

```
<h1 id="title">Hello</h1>
```

JavaScript

```
document.getElementById("title").innerHTML="Welcome";
```

Change Style

```
document.getElementById("title").style.color="blue";
```

Lesson 10: Events

Button

```
<button id="btn">Click</button>
```

JavaScript

```
document.getElementById("btn").addEventListener("click",function(){
alert("Button Clicked");
});
```

Lesson 11: Asynchronous JavaScript

Promises

```
const promise=new Promise((resolve,reject)=>{
resolve("Done");
});
```

Async Await

```
async function getData(){
let response=await fetch("https://jsonplaceholder.typicode.com/posts/1");
let data=await response.json();

console.log(data);
}

getData();
```

Mini Project

Student Grade Calculator

Requirements

- Input student marks
- Calculate percentage
- Assign grade
- Display result on webpage

Suggested Concepts

- Variables
- Functions
- Conditions
- DOM
- Events

Final Quiz

1. What is JavaScript used for?
2. Difference between let and const?
3. What is the purpose of functions?
4. Explain arrays and objects.
5. What is the DOM?
6. Difference between == and ===?
7. What are promises?
8. What is async/await?
9. What is event handling?
10. Build a simple calculator using JavaScript.

Course Summary

Congratulations!

You have learned:

- JavaScript basics
- Variables
- Operators
- Conditions
- Loops
- Functions

- Arrays
- Objects
- DOM Manipulation
- Events
- Asynchronous JavaScript

You are now ready to build interactive web applications and continue learning advanced JavaScript concepts such as ES6 modules, classes, APIs, and frameworks like React.